# MATTHEW C. PHARR

matthew.pharr@iter.org; +33 7 72 59 67 14
6 Rue Chastel, Aix-en-Provence 13100, France

## EDUCATION

*Columbia University*, New York, NY
Ph.D. Applied Physics, 2026
M.Phil. Applied Physics, 2025
M.S. Applied Physics, 2023

*Rensselaer Polytechnic Institute*, Troy, NY
B.S. Physics & Mathematics, 2021
*Summa Cum Laude*

## PUBLICATIONS

- **M. Pharr**, N. C. Logan, J. K. Park, and C. Paz-Soldan, *Resonant and non-resonant error field effects in drift-kinetically perturbed tokamak equilibria*, Manuscript in preparation (2026).
- **M. Pharr**, E. Bursch, N. C. Logan, P. Lunia, J. K. Park, and C. Paz-Soldan, *Coordinate-invariant flux-surface Fourier analysis in tokamaks*, Phys. Plasmas 33, 092503 (2026).
- **M. Pharr**, N. C. Logan, J. K. Park, and C. Paz-Soldan, *Quantifying resonant drive in resistive perturbed tokamak equilibria*, Nucl. Fusion 66, 086035 (2026).
- **M. Pharr**, N. C. Logan, C. Paz-Soldan, J. K. Park, and C. J. Hansen, *Error field predictability and consequences for ITER*, Nucl. Fusion 64, 126025 (2024).
- F. Ebrahimi and **M. Pharr**, *A nonlocal magneto-curvature instability in a differentially rotating disk*, Astrophys. J. 936, 145 (2022).

- N. C. Logan et al., *Unification and optimization of the Generalized Perturbed Equilibrium Code*, Manuscript in preparation (2026).
- S. Weber et al., *Fast and numerically robust error-field response from matrix Riccati integration*, Manuscript in preparation (2026).
- E. Bursch et al., *Improved $n = 1$ empirical error field penetration threshold scaling with Ohmic and L-mode conventional tokamak plasma discharges*, Plasma Phys. Control. Fusion 68, 105004 (2026).
- S. Freiberger et al., *Compact experimental negative triangularity reactor (CENTAUR): a design study for a compact, affordable breakeven tokamak*, Plasma Phys. Control. Fusion 68, 095025 (2026).
- N. C. Logan et al., *SPARC tokamak error field expectations and physics-based correction coil design*, Nucl. Fusion 66, 066003 (2026).
- X. Bai et al., *Time variation of error field correction in ITER*, Plasma Phys. Control. Fusion 67, 095013 (2025).
- S. Guizzo et al., *Electromagnetic system conceptual design for a negative triangularity tokamak*, Fusion Eng. Des. 219, 115257 (2025).
- The MANTA Collaboration et al., *MANTA: a negative-triangularity NASEM-compliant fusion pilot plant*, Plasma Phys. Control. Fusion 66, 105006 (2024).
- C. J. Hansen et al., *TokaMaker: an open-source time-dependent Grad-Shafranov tool for the design and modeling of axisymmetric fusion devices*, Comput. Phys. Commun. 298, 109111 (2024).
- C. Paz-Soldan et al., *Simultaneous access to high normalized density, current, pressure, and confinement in strongly-shaped diverted negative triangularity plasmas*, Nucl. Fusion 64, 094002 (2024).
- C. T. Holcomb et al., *DIII-D research to provide solutions for ITER and fusion energy*, Nucl. Fusion 64, 112003 (2024).

## RESEARCH & PROJECTS

*Monaco Postdoctoral Fellow, ITER Organization* October 2026 – Present
Science Division
Supervisor: Dr. Simon Mcintosh.

- Apply linear MHD to set installation tolerances for PF coils.
- Develop a holistic resonant and nonresonant error field correction strategy for ITER.

*Graduate Research Assistant, Columbia University* August 2021 – August 2026
Department of Applied Physics and Applied Mathematics, New York, NY
Advisors: Dr. Carlos Paz-Soldan, Dr. Nikolas Logan.

- Conducted research in theoretical and computational plasma physics, focusing on 3D perturbations of tokamak equilibria, error field effects and their induced risks, and MHD stability.
- Created an error field source model for ITER, and used it to predict the risk of error field penetration in ITER's baseline scenario. Used this model to propose a novel method for ensuring extrapolability of error field measurements from low to high beta.
- Developed methods for quantifying the resonant drive of magnetic islands in non-ideal perturbed tokamak equilibria.
- Contributed to the MANTA project, a joint Columbia/MIT negative triangularity pilot plant design study. Worked on integrated modeling of plasma equilibria, transport, and 0D systems modeling.
- Developed and applied computational tools, including: GPEC, a fast linear perturbed equilibrium suite, including sub-modules DCON and its asymptotic matching code, RDCON/RMATCH, as well as its Julia-based modernization; TokaMaker, a flexible, open-source Grad-Shafranov solver for tokamak and dipole equilibria; OpenPOPCON, an open-source 0D tokamak systems code that has been used for scoping the MANTA and CENTAUR tokamak designs, as well as pedagogically and in consultation with fusion startups.
- Worked for the SPARC MHD team, including maintaining and developing its total error field source model, giving feedback on iterative coil design, and predicting time-dependent risk of error field penetration. Developed a module for the POPSIM code, a plasma control system simulator, to include effects of time-dependent error fields and their driven tearing modes and penetration risks.
- Attended MIT CPS-FR Computational Summer School, 2023.

*SULI Intern, Princeton Plasma Physics Laboratory* Fall 2020 – Summer 2021
Theory Department, Princeton, NJ
Supervisor: Dr. Fatima Ebrahimi.

- Studied large-scale magnetic field growth and stability in Hall-MHD simulations of quasi-Keplerian plasma flows, with applications to astrophysical accretion disks.
- Used the extended MHD code NIMROD to perform linear and nonlinear simulations of magnetized plasmas in cylindrical geometry.
- Discovered a new family of more global solutions to the flow-driven magnetorotational instability (MRI) through a hierarchy of models, from a new fast linear eigenvalue solver to full 3D nonlinear simulations on large-scale computing clusters.

*Undergraduate Research Assistant, Rensselaer Polytechnic Institute* Summer 2020 – Spring 2021
Department of Mathematical Sciences, Troy, NY
Supervisors: Dr. Peter Kramer, Dr. Scott Forth.

- Studied the application of stochastic methods to the modeling of complex biological systems, including meiotic spindle assembly.
- Used Python to implement a computational statistical model of microtubule dynamics, and performed data analysis and visualization.

## PRESENTATIONS & CONFERENCE PROCEEDINGS

- **M. Pharr** et al., *Resistive and kinetic implications for ITER error field correction*, 47th ITPEA MDC Meeting, Seoul, Republic of Korea (2026).
- **M. Pharr** et al., *Resistive modifications to ideal error field sensitivity in tokamaks*, 30th Workshop on MHD Stability Control, San Diego, CA, USA (2026).
- **M. Pharr** et al., *Resistive and kinetic modifications to ideal error field sensitivity in tokamaks*, US-Japan MHD Workshop, Toki, Japan (2026).
- **M. Pharr** et al., *Quantifying the resonant drive for magnetic islands in perturbed ideal, resistive, and kinetic MHD equilibria*, 67th Annual Meeting of the APS Division of Plasma Physics, Long Beach, CA, USA (2025), abstract GO05.00007.

- **M. Pharr** et al., *Quantifying the resonant drive for magnetic islands in perturbed ideal, resistive, and kinetic MHD equilibria*, 29th US-Japan Joint Workshop on MHD Stability Control, Princeton, NJ, USA (2025).
- **M. Pharr** et al., *Expected error fields in ITER: a full-device source model and strategies for stable operation*, ITER International Summer School, Aix-en-Provence, France (2025).
- **M. Pharr** et al., *Expected error fields in ITER: a full-device source model and strategies for safe operation*, 66th Annual Meeting of the APS Division of Plasma Physics, Atlanta, GA, USA (2024), abstract UO06.00014.
- **M. Pharr** et al., *Error field predictability and consequences for ITER*, 65th Annual Meeting of the APS Division of Plasma Physics, Denver, CO, USA (2023), abstract PP11.00050.
- **M. Pharr** and F. Ebrahimi, *A nonlocal curvature-driven flow-shear instability in low-field plasmas*, Sherwood Fusion Theory Conference, Knoxville, TN, USA (2023).
- **M. Pharr** et al., *Error field source identification in early ITER plasmas*, 64th Annual Meeting of the APS Division of Plasma Physics, Spokane, WA, USA (2022), abstract PP11.00043.
- **M. Pharr**, F. Ebrahimi, and E. Blackman, *Large scale magnetic field growth and stability in Hall-MHD simulations of quasi-Keplerian flows*, 63rd Annual Meeting of the APS Division of Plasma Physics, Pittsburgh, PA, USA (2021), abstract BO06.00003.

## HONORS & AWARDS

*ORFEAS Fusion Design Contest Award*; Columbia University 2022
For contributions to the Columbia/MIT team's project on negative triangularity reactor pilot plant design scoping. Awarded $22,000 as a group.

*Max Hirsch Prize in Mathematics*; Rensselaer Polytechnic Institute 2021
This prize is awarded to a Senior in the Department of Mathematical Sciences who has demonstrated outstanding ability in his or her academic work and also gives promise of outstanding success in a career in mathematical sciences.

*J. Lawrence and Gertrude Katz Award in Physics*; Rensselaer Polytechnic Institute 2021
This award is presented to the student selected as the outstanding graduating senior receiving a Bachelor of Science in Physics.

ΣΠΣ, Undergraduate Physics Honor Society 2021

Honorable Mention for Research Paper, Mathematical Contest in Modeling 2020

Rensselaer Archimedian Society (4.0/4.0 GPA award) 2019

Rensselaer Leadership Award 2018

## PEDAGOGY

*Adjunct Instructor, Intro Physics Lab Sequence* Fall 2022 – Spring 2026
Barnard College of Columbia University, Physics and Astronomy, New York, NY

*Senior Graduate Mentor, Fusion Reactor Design* Fall 2024
Columbia University, Applied Physics and Applied Mathematics, New York, NY

*Graduate Teaching Assistant, Complex Analysis/Linear Algebra* Fall 2021 – Spring 2022
Columbia University, Applied Physics and Applied Mathematics, New York, NY

*Undergraduate Facilitator* Fall 2019 – Summer 2020, Spring 2021
Rensselaer Polytechnic Institute, Physics Department, Troy, NY

*I-PERSIST Mentor* Fall 2019
Rensselaer Polytechnic Institute, Physics Department, Troy, NY

*ALAC Introductory Physics Tutor* Fall 2019 – Spring 2020
Rensselaer Polytechnic Institute, Physics Department, Troy, NY

*Private Tutor, Math/Physics* Spring 2018 – Spring 2020
Self-employed. Provided 2–4 hours per week of university-level tutoring.

## SKILLS

*Languages and tools:*
Python, Julia, Fortran 77/90/95, MATLAB, Linux, Git, LaTeX, OpenMP, MPI
Amazon Web Services, Oracle Cloud, Google Cloud
Fortran and C++ graphical / memory debugging tools
Linear/nonlinear MHD simulation and analysis with DCON, GPEC, NIMROD

*General/practical skills:*
High-performance computing
Analytic and numerical modeling of plasmas
Application of neural networks and machine learning to physics problems
Use of vacuum technology and other plasma-physics lab equipment
Use of high-voltage lab equipment
IT and computer networking
Native English, B1 French, introductory Italian, introductory Mandarin